HiLiftAeroML Dataset Splits

Deterministic train/val/test splits for the HiLiftAeroML dataset. Covers the **1800 complete LHC cases**: 180 geometries × 10 angles of attack (4-22 deg).

Splits at a glance

Split	Type	Train	Val	Test	What it tests
full	In-dist	1260	180	360	Baseline: random case-level split
medium	In-dist	510	180	360	Intermediate data efficiency between full and scarce
scarce	In-dist	210	180	360	Data efficiency (1/6 of full training data)
super_scarce	In-dist	35	180	360	Extreme data efficiency (1/36 of full training data)
geometry	In-dist	1260	180	360	Generalization to unseen geometries
geometry_medium	In-dist	510	180	360	Unseen-geometry generalization from 51 training geometries
geometry_scarce	In-dist	210	180	360	Unseen-geometry generalization from 21 training geometries
geometry_super_scarce	In-dist	40	180	360	Unseen-geometry generalization from 4 training geometries
single_aoa_4	In-dist	126	18	36	Geometry generalization at 4 deg (pre-stall)
single_aoa_12	In-dist	126	18	36	Geometry generalization at 12 deg (mid-range)
single_aoa_22	In-dist	126	18	36	Geometry generalization at 22 deg (post-stall)
aoa	OOD	788	112	900	AoA extrapolation: low AoA → high AoA
deflection	OOD	1260	180	360	Geometry extrapolation: low → high deflection
stall	OOD	942	135	723	Flow regime: pre-stall → post-stall

Difficulty and data-efficiency ladders:

- Case-level data efficiency: full < medium < scarce < super_scarce (1260 / 510 / 210 / 35 training cases, same val/test)
- Geometry data efficiency: geometry < geometry_medium < geometry_scarce < geometry_super_scarce (126 / 51 / 21 / 4 training geometries, all 10 AoAs per geometry, same val/test)
- Geometry generalization: full < geometry < deflection

Which split should I use?

- **Simple baseline** → full
- **Geometry generalization** (primary ML challenge) → geometry
- **Hardest OOD challenge** → deflection or stall
- **AoA extrapolation** → aoa
- **Case-level data efficiency study** → compare super_scarce, scarce, medium, and full (same val/test)
- **Geometry data efficiency study** → compare geometry_super_scarce, geometry_scarce, geometry_medium, and geometry (same held-out geometries)
- **Single-AoA evaluation** → single_aoa_4, single_aoa_12, single_aoa_22

Design principles

- **Val is always in-distribution with train.** For OOD splits, the val set is drawn from the training-side population, never from the OOD test region. Hyperparameter tuning never sees out-of-distribution data.
- **Consistent ratios.** Splits with random/ranked geometry selection use 80/20 test fraction (70/10/20 overall). The aoa and stall splits have ratios dictated by their physics-based boundaries.
- **Shared held-out geometries.** The geometry* and single_aoa_* splits use the same 18 val and 36 test geometries, enabling direct comparison across training-set sizes and AoA regimes.
- **Test set integrity.** The test set should not be used to inform any training or hyperparameter decisions.

Split details

full

Random 70/10/20 partition of all 1800 cases. A given geometry may appear at different AoA values in train, val, and test. The model may have seen the same geometry at other angles of attack.

medium, scarce, and super_scarce

Same val/test as full, but with nested random subsamples of full_train. medium contains 510 cases, near the geometric midpoint between full (1260) and scarce (210); scarce uses 1/6 of full_train (210 cases), and super_scarce uses 1/36 (35 cases). A single deterministic priority ordering defines every level, so super_scarce_train ⊂ scarce_train ⊂ medium_train ⊂ full_train. The added midpoint resolves the previously six-fold jump between full and scarce while leaving every published assignment unchanged.

geometry

Splits the 180 geometries into 126 train, 18 val, and 36 test (all 10 AoA values per geometry). Unlike full, the model never sees a test geometry during training - not even at a different AoA - so it must generalize to entirely new shapes. Val and test geometries are randomly selected from the same LHC distribution as training.

geometry_medium, geometry_scarce, and geometry_super_scarce

Nested data-efficiency variants of geometry. They retain the exact same 18 validation and 36 test geometries as geometry, but reduce the training population to 51, 21, and 4 complete geometries, respectively. Every selected training geometry retains all 10 AoAs, giving 510, 210, and 40 training cases. One deterministic geometry ordering defines the nested relationship geometry_super_scarce_train ⊂ geometry_scarce_train ⊂ geometry_medium_train ⊂ geometry_train. Keeping complete AoA trajectories isolates the effect of geometry coverage instead of confounding it with incomplete operating-condition coverage.

aoa

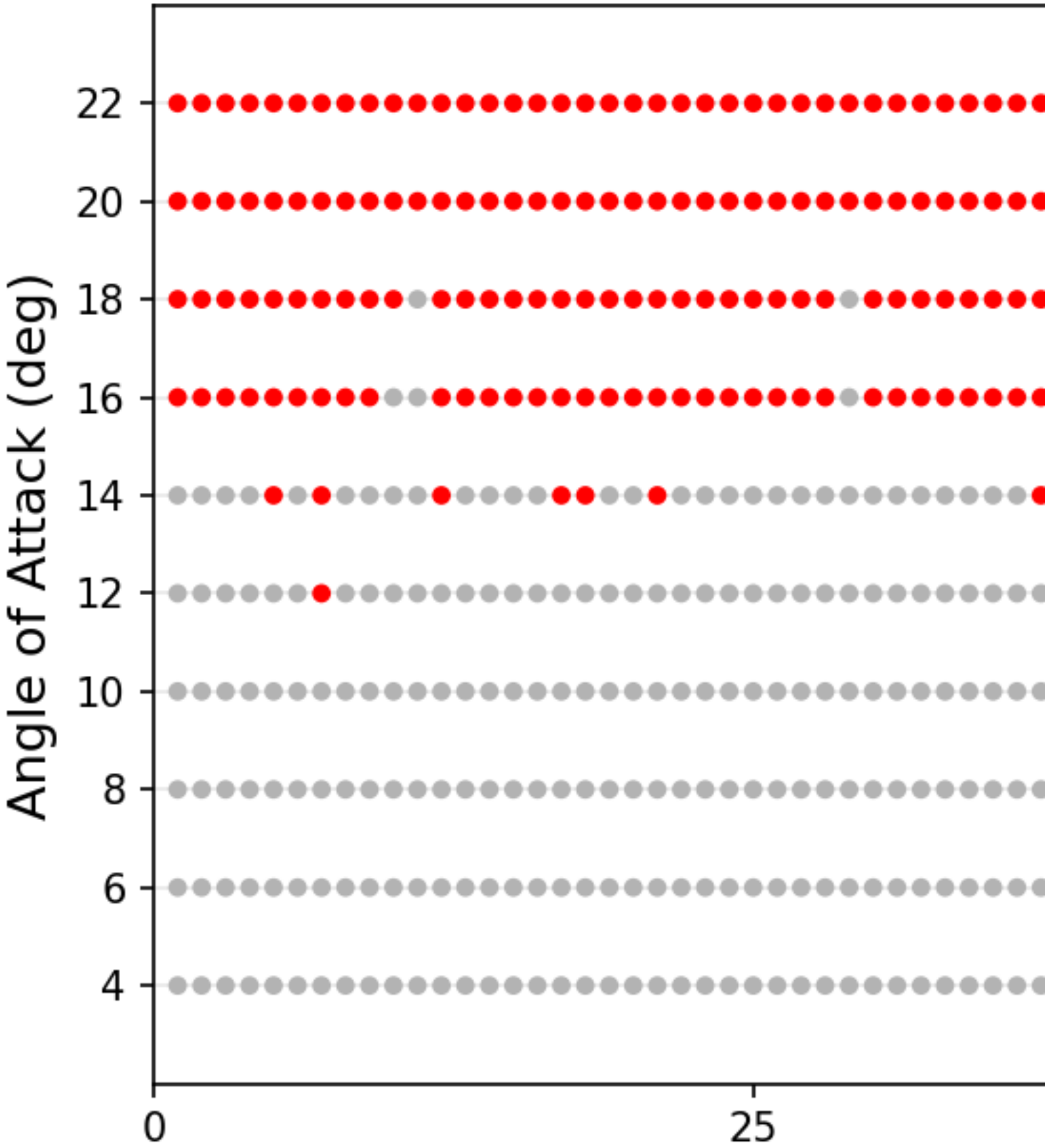
Trains and validates on AoA ≤ 12 (pre-stall regime); tests on AoA ≥ 14. All 180 geometries appear in all three sets. The 12/14 deg cutoff is physics-motivated: around 14-16 deg, the dominant aerodynamic sensitivity shifts from flap deflection to slat deflection, and most geometries begin to stall. Val is a random 1/8 of the pre-stall pool.

deflection

Sorts geometries by mean deflection angle (average of IB/OB flap and slat deflections). Trains and validates on the bottom 80% (low deflection); tests on the top 20% (most aggressive high-lift settings). Val is a random 1/8 of the low-deflection pool. This is a harder variant of geometry - both hold out entire geometries, but deflection selects adversarially from the extremes of the design space rather than randomly.

stall

For each geometry, fits a cubic spline to CL(α) and identifies the first AoA where dCL/dα ≤ 0 (stall onset). Everything from that AoA onward is post-stall. Trains and validates on pre-stall cases; tests on all post-stall cases. Val is a random 1/8 of the pre-stall pool. All 180 geometries exhibit stall - the majority (100) at 16 deg, with a secondary cluster (44) at 14 deg.


Angle of Attack (deg)
22
20
18
16
14
12
10
8
6
4
0
25

single_aoa_4, single_aoa_12, single_aoa_22

Per-AoA variants of geometry: same 126/18/36 geometry split, restricted to a single angle of attack. AoA 4 (pre-stall), 12 (mid-range), and 22 (post-stall) represent three distinct flow regimes.

Regenerating

```
uv run splits/generate_splits.py
```

All 14 split families are deterministic (fixed seed). The script reads CL data from the dataset for the stall split and geometry parameters for the deflection split.

Manifest format

```
{
"full_train": ["geo_LHC001_AoA_4", ...],
"full_val": [...],
"full_test": [...],
...
}
```

Case IDs match on-disk directory names, sorted by geometry number then AoA.